

INTERNSHIP REPORT

A report submitted in partial fulfillment of the requirements for the Award of Degree

of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

By

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INTERNSHIP CERTIFICATE

This is to certify that the “INTERNSHIP REPORT” submitted by Dhaani Bahl 22BCE11071 for the internship done by her during the academic year 2025-2026 from 06-02-2025 to 23-05-2025 (3 Months), is the partial fulfillment of the requirements for the awarding the degree of B. Tech – CSE.

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CERTIFICATION



1. DevOps Fundamentals



2. Agile Methodologies Version: 1



3. Agile Methodologies Version: 2



4. Machine Learning

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to VIT Bhopal University for providing the infrastructure and support that made this internship opportunity possible. I am especially thankful to the Placement Office for their continuous efforts in connecting students with valuable industry-oriented learning programs. Through their initiative, I had the opportunity to pursue the professional courses **DevOps Fundamentals and Agile Methodologies by IBM** and **Machine learning by SmartBridge in collaboration with Google for Developers**, which significantly enhanced my technical understanding and practical exposure to modern technologies. I would also like to sincerely thank IBM and SmartBridge for designing structured learning modules that helped bridge the gap between academic knowledge and real-world industry applications. Special appreciation goes to the Programme Chair and faculty members for offering such relevant courses aligned with current technological trends and for providing constant guidance throughout the duration of the program. Their encouragement and academic support during these intensive months of learning played a crucial role in strengthening my technical and professional skills. This experience has greatly contributed to my growth as a student and has provided valuable insight into practical industry tools and workflows. Finally, I extend my sincere thanks to all mentors and administrative staff whose efforts behind the scenes ensured a smooth and enriching internship experience.

ABSTRACT

This report outlines the comprehensive learning experience and skill development achieved through the successful completion of two professional certification courses: **DevOps Fundamentals and Agile Methodologies** offered by IBM, and **Machine Learning** conducted by SmartBridge in collaboration with Google for Developers. These courses were undertaken with the objective of gaining both theoretical understanding and practical exposure to modern software engineering practices and data-driven technologies that are shaping the current and future landscape of the IT industry.

The **DevOps Fundamentals and Agile Methodologies** course provided an in-depth understanding of the principles and practices that enable organizations to deliver high-quality software efficiently and reliably. It began with an introduction to the traditional software development lifecycle and its limitations, followed by the evolution of DevOps as a solution to bridge the gap between development and operations teams. Key concepts such as Continuous Integration (CI), Continuous Delivery (CD), and Continuous Deployment were explored in detail, emphasizing how automation plays a critical role in reducing manual effort, minimizing errors, and accelerating the development process.

The course also highlighted the importance of collaboration, communication, and shared responsibility among cross-functional teams. Agile methodologies, including Scrum and Kanban, were discussed as frameworks that promote iterative development, flexibility, and rapid feedback. Concepts such as sprint planning, daily stand-ups, backlog management, and retrospectives were introduced to demonstrate how teams can continuously improve their processes and deliver value to stakeholders. Additionally, exposure to tools and practices related to version control, build automation, testing, and monitoring provided a holistic view of the DevOps lifecycle, which includes planning, coding, building, testing, releasing, deploying, operating, and monitoring applications in real-world scenarios.

In parallel, the **Machine Learning** course offered a strong foundation in the concepts and applications of data analysis and intelligent systems. The course began with an introduction to the significance of machine learning in solving real-world problems across domains such as healthcare, finance, marketing, and technology. It covered the fundamental types of machine learning, including supervised learning, unsupervised learning, and an overview of reinforcement learning. Key topics such as data preprocessing, data cleaning, handling missing values, and feature engineering were emphasized as critical steps in preparing data for model training.

Furthermore, the course provided hands-on experience in building, training, and evaluating machine learning models using Python and relevant libraries. Algorithms such as linear regression, logistic regression, decision trees, and clustering techniques were introduced to demonstrate different approaches to prediction and pattern recognition. Model evaluation techniques, including accuracy, precision, recall, and validation methods, were also discussed to ensure the reliability and performance of the developed models.

Overall, the completion of these two courses has contributed to the development of a well-rounded skill set that combines software development efficiency with data-driven intelligence. While the DevOps and Agile training strengthened the understanding of collaborative development, automation, and efficient project management, the Machine Learning course enhanced the ability to analyze data, build predictive models, and derive meaningful insights.

METHODOLOGIES LEARNED

Methodology: DevOps and Agile methodologies

The **DevOps Fundamentals and Agile Methodologies** course by IBM introduced several key methodologies that are widely used in modern software development to improve efficiency, collaboration, and product quality.

One of the primary methodologies learned was **Agile**, which focuses on iterative and incremental development. Instead of building an entire product at once, Agile promotes breaking the project into smaller, manageable units called iterations or sprints. This approach allows teams to adapt quickly to changing requirements, gather continuous feedback, and deliver functional components at regular intervals. Agile emphasizes customer collaboration, flexibility, and continuous improvement over rigid planning.

Within Agile, specific frameworks such as **Scrum** were explored. Scrum provides a structured approach to managing work through defined roles, events, and artifacts. Key roles include the Product Owner, Scrum Master, and Development Team, each responsible for different aspects of the project. Important practices such as sprint planning, daily stand-up meetings, sprint reviews, and retrospectives help ensure transparency, accountability, and continuous progress. This framework enhances team coordination and helps in identifying and resolving issues early.

Another methodology covered was **Kanban**, which focuses on visualizing workflows and optimizing the flow of tasks. Using tools like boards and cards, Kanban helps teams track progress, limit work in progress (WIP), and identify bottlenecks in real time. Unlike Scrum, Kanban is more flexible and does not require fixed-length iterations, making it suitable for continuous delivery environments.

The course also emphasized DevOps methodology, which integrates development and operations to create a seamless software delivery pipeline. DevOps focuses on automation, collaboration, and continuous monitoring throughout the software lifecycle. Practices such as Continuous Integration (CI) and Continuous Delivery (CD) were key highlights, where code changes are automatically built, tested, and prepared for deployment. This reduces manual intervention, minimizes errors, and accelerates the release cycle.

Additionally, the concept of Infrastructure as Code (IaC) was introduced, where infrastructure setup and configuration are managed using code rather than manual processes. This ensures consistency, scalability, and easier maintenance of systems. Continuous Monitoring and Feedback was another important aspect, enabling teams to track application performance and quickly respond to issues.

Methodology: Machine Learning

The **Machine Learning** course conducted by SmartBridge in collaboration with Google for Developers introduced fundamental methodologies used to build data-driven and intelligent systems. The course focused on the complete machine learning pipeline, starting from data collection and preprocessing to model development and evaluation. A strong emphasis was placed on preparing data through techniques such as data cleaning, handling missing values, normalization, and feature selection, as these steps significantly influence model accuracy and performance.

One of the core methodologies covered was supervised learning, where models are trained on labeled datasets to perform tasks such as prediction and classification. Algorithms like linear regression and classification models helped in understanding how relationships between variables are identified and used for decision-making. In addition, unsupervised learning was explored, where models analyze unlabeled data to detect hidden patterns or groupings, such as clustering. These approaches provided insight into different ways of extracting meaningful information from datasets.

Another important methodology introduced was the concept of model training and testing workflows, where datasets are split into training and testing sets to evaluate model performance on unseen data. This ensures that the model generalizes well rather than simply memorizing the training data. The importance of feature engineering was also highlighted, where relevant features are selected or transformed to improve model efficiency and accuracy.

The course further emphasized model evaluation and optimization techniques, including the use of metrics such as accuracy, precision, recall, and validation strategies. Concepts like bias and variance were discussed to understand common challenges such as underfitting and overfitting. Techniques to improve model performance, including tuning parameters and selecting appropriate algorithms, were also introduced.

Additionally, the methodology of iterative improvement in machine learning was highlighted, where models are continuously refined based on performance results and feedback. This process involves experimenting with different algorithms, adjusting parameters, and improving data quality to achieve better outcomes. Overall, these methodologies provided a structured understanding of how machine learning systems are designed, developed, evaluated, and improved to solve real-world problems effectively.

LEARNING OBJECTIVES

DevOps Fundamentals and Agile Methodologies (IBM)

- To understand the core concepts of DevOps, including continuous integration, continuous delivery (CI/CD), and automation in software development.
- To learn how development and operations teams collaborate to improve software quality and deployment speed.
- To gain practical knowledge of tools used in DevOps pipelines, such as version control systems, build tools, and deployment workflows.
- To understand Agile methodologies like Scrum and Kanban, and how they help in iterative and efficient project management.
- To develop skills in planning, tracking, and managing tasks using Agile practices such as sprints, stand-ups, and backlog management.
- To learn how to improve software reliability and scalability through monitoring, testing, and continuous feedback.

Machine Learning (SmartBridge in collaboration with Google for Developers)

- To build a strong foundation in machine learning concepts, including supervised and unsupervised learning techniques.
- To understand the process of data preprocessing, feature selection, and model training using real-world datasets.
- To gain hands-on experience with Python libraries such as Pandas, NumPy, and Scikit-learn for implementing ML models.
- To learn how to evaluate model performance using metrics like accuracy, precision, recall, and confusion matrix.
- To explore different machine learning algorithms such as linear regression, decision trees, and clustering techniques.
- To understand how machine learning models can be applied to solve real-world problems and support data-driven decision-making.
- To develop the ability to interpret model results and communicate findings effectively.

WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

DevOps and Agile methodologies

Week	Topics Covered	Learning	Project Progress (CI/CD Pipeline)
Week 1	Introduction to DevOps & SDLC	Learned basics of DevOps, SDLC models, and need for DevOps	Planned project structure and understood pipeline workflow
Week 2	Version Control	Learned Adroit pro-learn commands, branching, and folder management	Set up folders and saved initial project code
Week 3	Continuous Integration (CI)	Understood CI concepts, automated builds, and testing	Implemented CI process to trigger builds on code commits
Week 4	Continuous Delivery/Deployment (CD)	Learned deployment strategies and automation tools	Added automated deployment stage to pipeline
Week 5	Agile Methodologies (Scrum, Kanban)	Learned sprint planning, stand-ups, and workflow management	Improved pipeline workflow using iterative approach
Week 6	Monitoring & Optimization	Learned monitoring, logging, and performance improvement	Finalized fully functional CI/CD pipeline with automation and testing

Machine Learning

Week	Topics Covered	Learning	Project Progress (CI/CD Pipeline)
Week 1	Introduction to Machine Learning	Learned types of ML, real-world applications	Understood dataset and defined project problem
Week 2	Data Preprocessing	Data cleaning, handling missing values, feature selection	Cleaned dataset and prepared it for modeling
Week 3	Supervised Learning	Learned regression and classification algorithms	Implemented initial models (regression/classification)
Week 4	Unsupervised Learning	Learned clustering and pattern recognition	Explored patterns in visa dataset
Week 5	Model Evaluation	Accuracy, precision, recall, validation techniques	Evaluated model performance and improved accuracy
Week 6	Model Optimization & Finalization	Learned tuning and optimization techniques	Finalized H1-B visa prediction model and tested results

SKILLS ACQUIRED

The completion of the **DevOps Fundamentals and Agile Methodologies** course by IBM and the **Machine Learning** course by SmartBridge in collaboration with Google for Developers contributed significantly to the development of a diverse and industry-relevant skill set. These skills span across software development practices, data analysis, problem-solving, and the ability to build scalable and intelligent systems.

One of the major skills acquired is proficiency in modern software development practices, particularly through the understanding of DevOps principles. This includes knowledge of Continuous Integration and Continuous Delivery (CI/CD), automation of workflows, and the ability to streamline the software development lifecycle. The course enhanced the ability to work with version control systems, manage code efficiently, and understand deployment strategies, which are essential for building reliable and scalable applications. Additionally, the exposure to Agile methodologies improved the ability to work in dynamic environments, adapt to changing requirements, and deliver results in an iterative and efficient manner.

Another key skill developed is collaboration and team-oriented problem solving. **Agile** frameworks such as **Scrum** and **Kanban** emphasize communication, coordination, and shared responsibility among team members. Through these methodologies, the ability to participate in structured workflows, manage tasks effectively, and contribute to team goals was strengthened. Skills such as planning, organizing, and reviewing work through sprints and feedback loops were also enhanced, which are highly valuable in real-world project environments.

In addition, model development and evaluation skills were acquired through hands-on experience with machine learning algorithms. This includes the ability to build predictive models, evaluate their performance using appropriate metrics, and improve them through optimization techniques. Understanding concepts such as bias, variance, and overfitting helped in developing more accurate and reliable models. These skills are essential for designing intelligent systems that can make informed decisions based on data.

Another important area of skill development is technical proficiency in tools and technologies. Both courses provided exposure to programming using **Python** and working with relevant libraries and tools used in machine learning and software development. Familiarity with such tools enhances the ability to implement concepts practically and work on real-world projects efficiently. It also builds confidence in handling complex systems and large datasets.

Furthermore, the courses helped in developing problem-solving and critical thinking abilities. Whether it is optimizing a software pipeline in DevOps or improving the accuracy of a machine learning model, both domains require a structured approach to identifying problems and implementing effective solutions.

Lastly, these courses contributed to the development of adaptability and continuous learning skills. The technology landscape is constantly evolving, and both DevOps and Machine Learning require staying updated with new tools, techniques, and best practices. Through this learning experience, the ability to quickly grasp new concepts and apply them effectively was enhanced.

RESULTS

The outcomes of the **DevOps Fundamentals and Agile Methodologies** course by IBM and the **Machine Learning** course by SmartBridge in collaboration with Google for Developers are clearly demonstrated through the successful development and implementation of practical, real-world projects in both domains. These results reflect not only theoretical understanding but also the ability to apply learned concepts effectively in solving real-world challenges.

As a key outcome of the DevOps course, a fully functional Continuous Integration and Continuous Delivery (CI/CD) pipeline was designed and implemented. The pipeline was structured to automate the entire software development lifecycle, starting from code integration to testing and deployment. Whenever changes were made to the codebase, the pipeline automatically triggered build processes, executed testing procedures, and ensured that only verified and error-free code progressed to the deployment stage. This significantly reduced manual effort, minimized the chances of human error, and improved overall development efficiency.

The implementation of this CI/CD pipeline demonstrated a strong understanding of core DevOps practices such as automation, version control, continuous testing, and monitoring. It also reflected the ability to integrate multiple stages of development into a seamless workflow. The pipeline ensured faster feedback cycles, allowing issues to be identified and resolved at an early stage. Additionally, the project highlighted the importance of collaboration between development and operations, aligning with the fundamental principles of DevOps.

In the Machine Learning domain, a significant outcome was the development of an H1-B visa prediction model, aimed at predicting the likelihood of visa approval based on historical data. The project involved working with structured datasets and applying a complete machine learning workflow. Initial stages included data preprocessing, where missing values were handled, irrelevant features were removed, and important variables were selected to improve model performance.

Following data preparation, multiple machine learning algorithms were explored and implemented to build predictive models. The model was trained using historical visa application data, enabling it to identify patterns and relationships between different factors influencing visa approval decisions. Evaluation techniques were applied to measure model performance using appropriate metrics, ensuring that the predictions were accurate and reliable. The process also involved refining and optimizing the model to achieve better performance and avoid issues such as overfitting.

The H1-B visa prediction model demonstrated the practical application of machine learning concepts in solving real-world problems. It showcased the ability to analyze complex datasets, extract meaningful insights, and develop predictive solutions that can assist in decision-making processes. The project also reflected strong analytical thinking, problem-solving ability, and technical proficiency in implementing machine learning techniques.

PROJECT IPLEMENTATION

The project implementation was carried out by applying concepts learned from both the DevOps Fundamentals and Agile Methodologies course by IBM and the Machine Learning course by SmartBridge in collaboration with Google for Developers. The implementation focused on developing practical solutions in both domains, integrating software development practices with data-driven modeling.

The DevOps implementation involved designing and developing a Continuous Integration and Continuous Delivery (CI/CD) pipeline. The process began with maintaining the codebase using version control systems, ensuring proper tracking of changes and collaboration. The pipeline was structured to automate key stages of the software development lifecycle, including code integration, building, testing, and deployment.

Whenever changes were made to the code, the pipeline automatically triggered build and testing processes. Only validated and error-free code was allowed to proceed to the deployment stage. This automation reduced manual intervention, minimized errors, and improved overall efficiency. Continuous monitoring and feedback mechanisms were also incorporated to ensure early detection and resolution of issues. Agile methodologies were followed throughout the implementation, where the project was divided into smaller tasks and executed in iterative cycles, allowing flexibility and continuous improvement.

In parallel, the Machine Learning implementation focused on developing an H1-B visa prediction model using historical data. The process started with data collection and preprocessing, where missing values were handled, irrelevant features were removed, and important variables were selected to improve model performance.

Following data preparation, exploratory data analysis was conducted to identify patterns and relationships within the dataset. Multiple machine learning algorithms were then implemented and trained on the processed data. The models were evaluated using appropriate performance metrics to ensure accuracy and reliability. Based on the evaluation results, the models were further optimized to improve prediction performance and reduce issues such as overfitting.

The final model was capable of predicting the likelihood of visa approval based on input parameters, demonstrating the practical application of machine learning techniques in solving real-world problems. Overall, the project successfully integrated DevOps practices and machine learning workflows, resulting in efficient development processes and effective predictive solutions.

DevOps and Agile methodologies

INTRODUCTION TO DEVOPS

IBM is a global leader in enterprise technology, known for its advancements in cloud computing, artificial intelligence, and software development solutions. The DevOps Fundamentals and Agile Methodologies course provided a comprehensive understanding of modern software development practices that emphasize speed, reliability, and collaboration.

DevOps is a methodology that bridges the gap between development and operations teams by promoting continuous integration, automated testing, and continuous delivery of software. It focuses on reducing development cycles while maintaining high-quality outputs. Alongside DevOps, Agile methodologies such as Scrum and Kanban were introduced, which help teams work in iterative cycles, adapt to changing requirements, and deliver incremental improvements efficiently.

This training provided both theoretical knowledge and practical insights into how real-world software systems are developed, tested, and deployed in an automated and collaborative environment.

CONCEPTS LEARNED

The course covered a wide range of essential DevOps and Agile concepts that are widely used in the industry.

One of the key concepts learned was Continuous Integration and Continuous Delivery (CI/CD). Continuous Integration involves regularly merging code changes into a shared repository, followed by automated build and testing processes. Continuous Delivery ensures that the validated code is always in a deployable state, enabling faster and more reliable releases.

Another important concept was version control systems, particularly Git. Version control helps in tracking code changes, managing multiple versions of a project, and enabling collaboration among team members without conflicts.

The course also emphasized automation, which plays a critical role in DevOps. Automation of tasks such as building, testing, and deployment reduces manual effort, minimizes errors, and improves consistency across the development lifecycle.

In addition, Agile methodologies were explored in detail. Scrum focuses on dividing work into sprints, conducting regular meetings (such as stand-ups), and continuously reviewing progress. Kanban, on the other hand, emphasizes visual task management and workflow optimization. These methodologies improve productivity, flexibility, and team coordination.

The concept of continuous monitoring and feedback was also introduced, which ensures that system performance is tracked and issues are identified early. This helps in maintaining system stability and improving overall software quality.

INTEGRATION TO PROJECT

The knowledge gained from the DevOps training was practically applied in the project by implementing a structured and automated development workflow.

The project began with setting up a version-controlled code repository, where all code changes were managed efficiently. This ensured proper tracking, easy rollback of changes, and seamless collaboration.

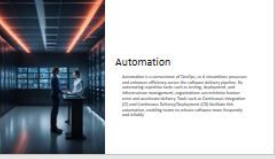
A CI/CD pipeline was then designed and implemented to automate the software development lifecycle. Whenever code was updated, the pipeline automatically triggered a sequence of processes including code integration, building, and testing. Automated tests were used to validate the functionality and detect errors at an early stage.

Only after successful testing, the code proceeded to the deployment phase, ensuring that only stable and error-free versions of the application were released. This significantly reduced manual intervention and improved the reliability of the system. Agile practices were followed throughout the project execution. The entire project was divided into smaller tasks, which were completed in iterative cycles or sprints. Regular evaluation and feedback allowed continuous improvement and quick adaptation to any changes or issues.

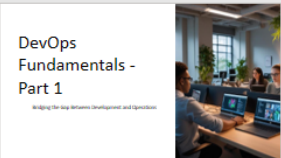
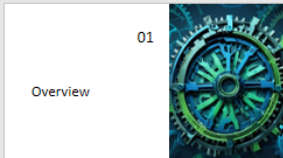
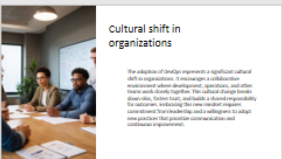
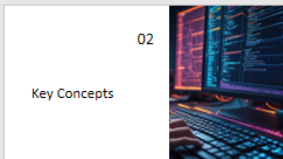
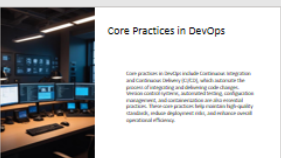
Additionally, basic monitoring and feedback mechanisms were incorporated to track performance and identify potential issues. This ensured that the system remained stable and efficient over time.

Overall, the implementation demonstrated how DevOps practices can streamline development processes, enhance collaboration, and improve the quality and speed of software delivery.

ASSIGNMENT 1: Getting to know the methodologies

 Agile and DevOps Introducing Agile Methodologies and its Relationship to DevOps Practices	 Introduction This presentation will cover key aspects of agile methodologies for providing key practices, and how they relate to DevOps practices	 01 Agile Overview	 Iterative Development Iterative Development is a core aspect of the agile methodology, allowing teams to work in repeated cycles of development, testing, and deployment. This approach enables teams to respond quickly to changes and deliver high-quality software incrementally. The iterative process typically involves frequent releases, continuous feedback, and the ability to pivot based on user requirements.
 Customer Collaboration Customer collaboration is a key aspect of agile methodologies, emphasizing the importance of working closely with customers throughout the development process. This approach ensures that the final product meets the customer's needs and expectations, leading to higher customer satisfaction and a product that is more likely to be successful in the market.	 Key Practices Key practices in agile methodologies include frequent releases, continuous feedback, and the ability to pivot based on user requirements. These practices are essential for ensuring that the final product meets the customer's needs and expectations, leading to higher customer satisfaction and a product that is more likely to be successful in the market.	 02 DevOps Overview	 Culture and Practices DevOps is a culture and a set of practices that emphasizes collaboration between development and operations teams. This approach aims to reduce the time and effort required to deliver new software, while also improving the quality and reliability of the software. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.
 Automation Automation is a key aspect of DevOps, enabling teams to automate repetitive tasks and streamline the development process. This approach reduces the risk of human error and improves the efficiency of the development process, leading to faster releases and higher quality software. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.	 Key Practices Key practices in DevOps include frequent releases, continuous feedback, and the ability to pivot based on user requirements. These practices are essential for ensuring that the final product meets the customer's needs and expectations, leading to higher customer satisfaction and a product that is more likely to be successful in the market.	 Conclusions In summary, agile methodologies and DevOps are essential for ensuring that the final product meets the customer's needs and expectations, leading to higher customer satisfaction and a product that is more likely to be successful in the market. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.	

ASSIGNMENT 2: Understanding the fundamentals

 DevOps Fundamentals - Part 1 Bridging the Gap Between Development and Operations	 Introduction DevOps is a cultural and technical movement that aims to bridge the gap between development and operations teams. This approach emphasizes collaboration, communication, and automation, leading to faster releases and higher quality software. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.	 01 Overview	 Definition of DevOps DevOps is a combination of cultural practices, automation, and tools that aim to bridge the gap between development and operations teams. This approach emphasizes collaboration, communication, and automation, leading to faster releases and higher quality software. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.
 Goals of DevOps The primary goal of DevOps is to improve the speed and quality of software delivery. This is achieved by automating repetitive tasks, streamlining the development process, and fostering collaboration between development and operations teams. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.	 Cultural shift in organizations The adoption of DevOps represents a significant cultural shift in organizations, emphasizing collaboration, communication, and automation. This approach aims to reduce the time and effort required to deliver new software, while also improving the quality and reliability of the software. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.	 02 Key Concepts	 DevOps Lifecycle phases The DevOps lifecycle consists of several phases, including planning, development, testing, deployment, and monitoring. Each phase is designed to ensure that the final product meets the customer's needs and expectations, leading to higher customer satisfaction and a product that is more likely to be successful in the market. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.
 Core Practices in DevOps Core practices in DevOps include frequent releases, continuous feedback, and the ability to pivot based on user requirements. These practices are essential for ensuring that the final product meets the customer's needs and expectations, leading to higher customer satisfaction and a product that is more likely to be successful in the market.	 Popular Tools in DevOps Popular tools in DevOps include Jenkins, Docker, Kubernetes, and Terraform. These tools are essential for automating repetitive tasks and streamlining the development process, leading to faster releases and higher quality software. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.	 Conclusions In summary, DevOps is a cultural and technical movement that aims to bridge the gap between development and operations teams. This approach emphasizes collaboration, communication, and automation, leading to faster releases and higher quality software. Key practices include frequent releases, continuous feedback, and the ability to pivot based on user requirements.	

Machine Learning using Python

INTRODUCTION TO MACHINE LEARNING

Machine Learning is a branch of artificial intelligence that enables systems to learn from data and make predictions or decisions without being explicitly programmed. It plays a crucial role in modern applications such as recommendation systems, fraud detection, and predictive analytics.

The Machine Learning training conducted by SmartBridge in collaboration with Google for Developers provided a strong foundation in both theoretical concepts and practical implementation. The training focused on building intelligent systems using real-world datasets and applying data-driven approaches to solve problems.

The implementation was carried out using Jupyter Notebook, an interactive platform widely used for data analysis, visualization, and model development. It allows step-by-step execution of code, making it easier to understand workflows and debug models efficiently.

CONCEPTS LEARNED

The course covered essential Machine Learning concepts required to build and evaluate predictive models.

One of the fundamental areas was data preprocessing, which involves cleaning the dataset by handling missing values, removing inconsistencies, and transforming raw data into a usable format. This step is critical as the quality of data directly affects model performance.

Another important concept was Exploratory Data Analysis (EDA). EDA helps in understanding the dataset by identifying patterns, trends, and relationships between variables using statistical methods and visualizations.

The course also focused on feature selection and feature engineering, where relevant variables are selected and transformed to improve model accuracy. Choosing the right features plays a major role in building efficient models.

A key component of the training was learning supervised machine learning algorithms, such as linear regression, decision trees, and classification models. These algorithms were used to train models on historical data and make predictions.

In addition, model evaluation techniques were studied, including metrics such as accuracy, precision, recall, and confusion matrix. These metrics help in measuring how well the model performs on unseen data.

The concepts of overfitting and underfitting were also introduced, along with techniques to optimize model performance and ensure generalization.

INTEGRATION TO PROJECT

The Machine Learning concepts were applied in the project through the development of an H1-B visa prediction model using Python.

The implementation began with data collection and preprocessing, where the dataset was cleaned by handling missing values, removing irrelevant attributes, and selecting important features. This ensured that the data was suitable for model training.

Next, Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset and identify key patterns influencing visa approval outcomes. Various visualizations were used to gain insights into relationships between variables.

Following this, multiple machine learning algorithms were implemented and trained on the dataset. Different models were tested to determine which algorithm provided the best performance.

The models were then evaluated using appropriate performance metrics, ensuring that predictions were accurate and reliable. Based on the evaluation results, the models were further refined and optimized to improve their efficiency and reduce overfitting.

The final model was capable of predicting the likelihood of visa approval based on input parameters, demonstrating the practical application of machine learning techniques in solving real-world problems.

Overall, the project showcased the complete machine learning workflow, from data preprocessing to model deployment, highlighting the effectiveness of data-driven decision-making.

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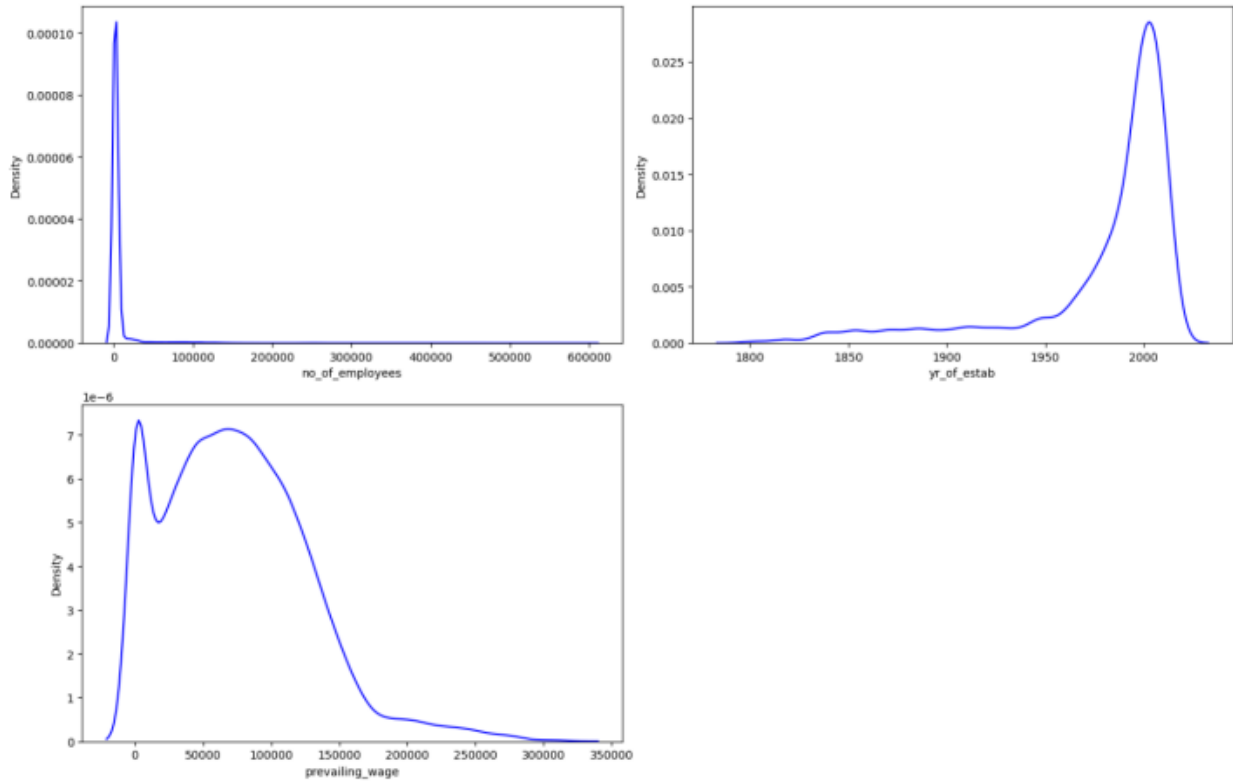
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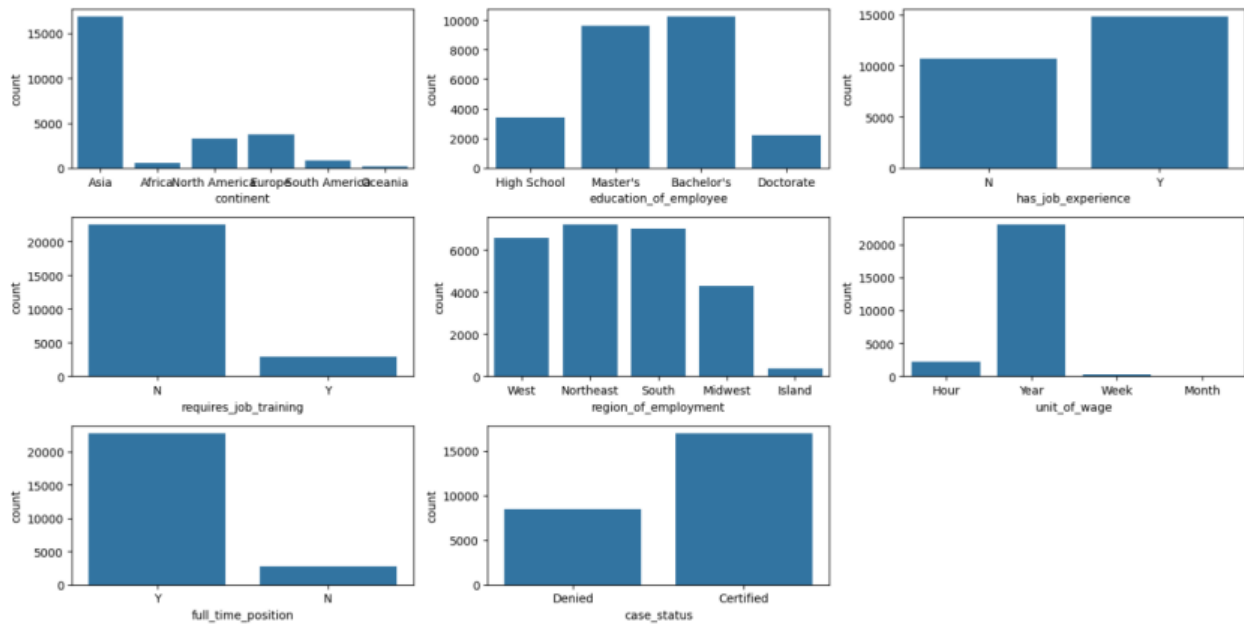
df.shape

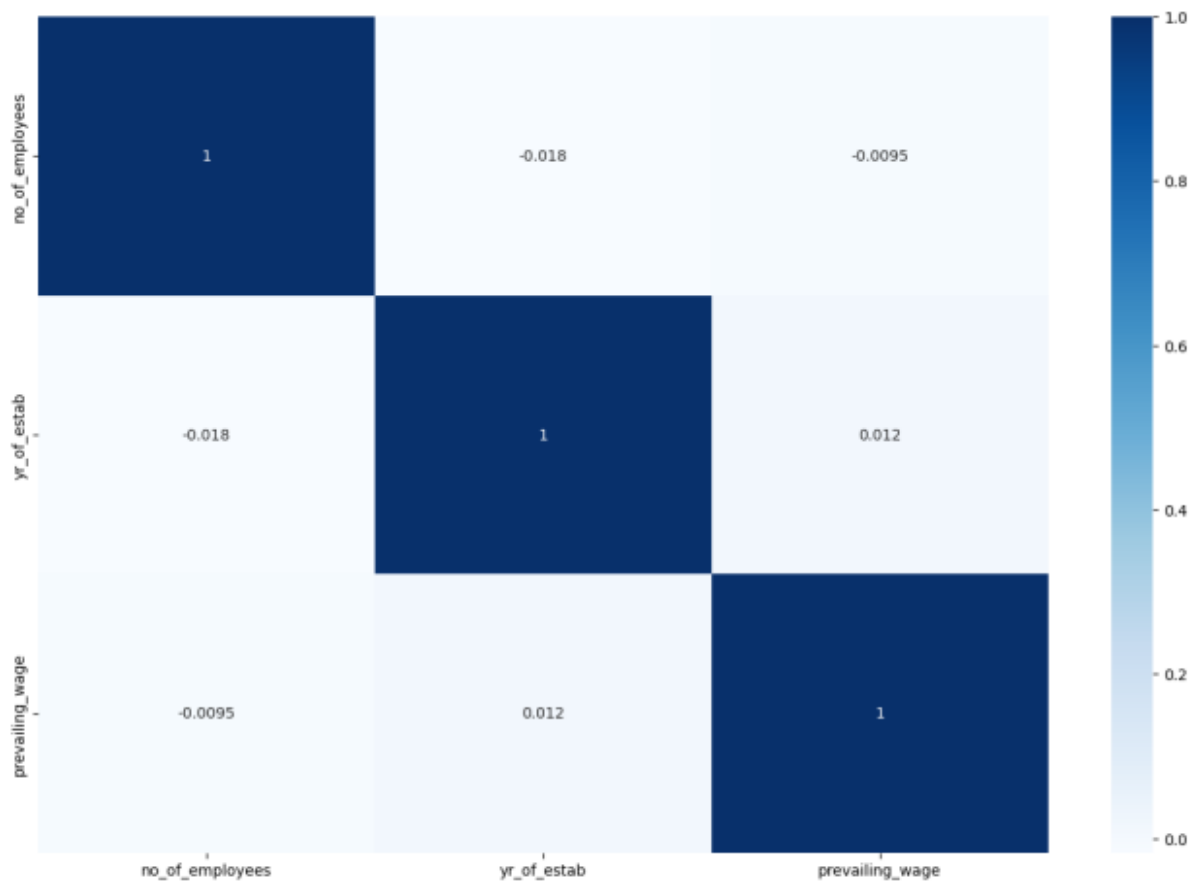
(25480, 12)

Univariate Analysis of Numerical Features



Univariate Analysis of Categorical Features

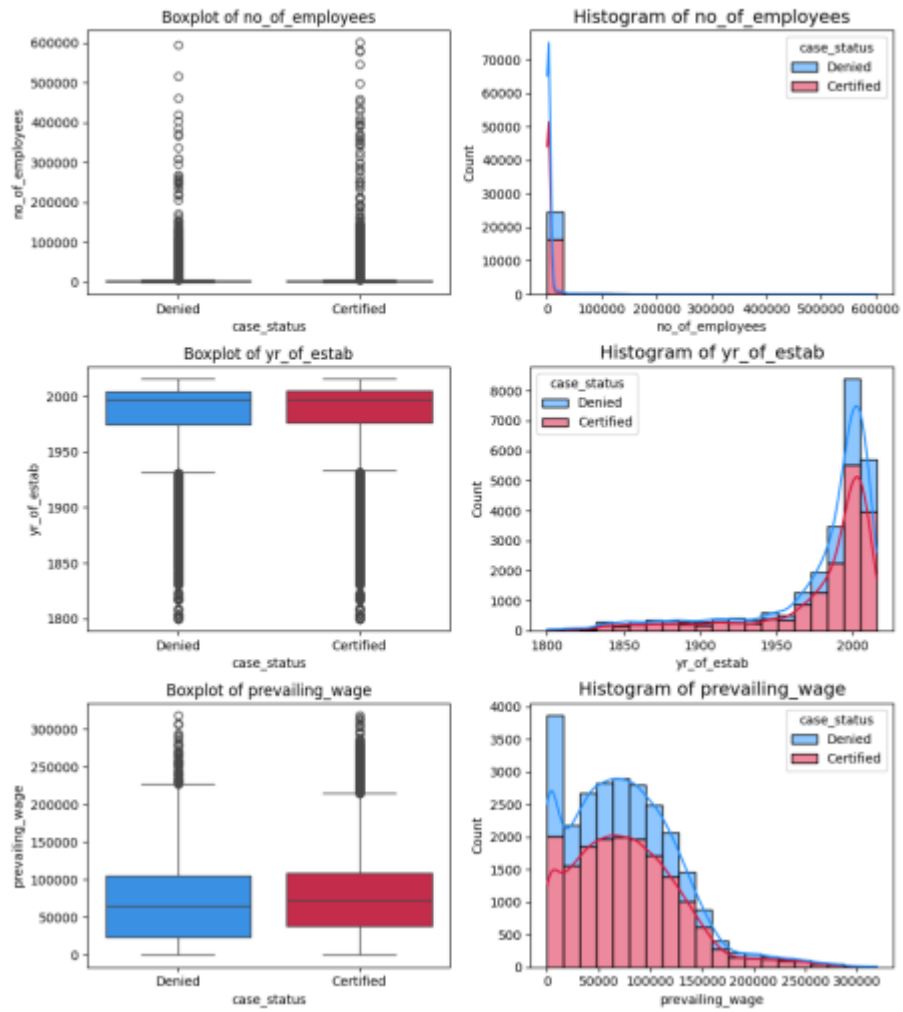


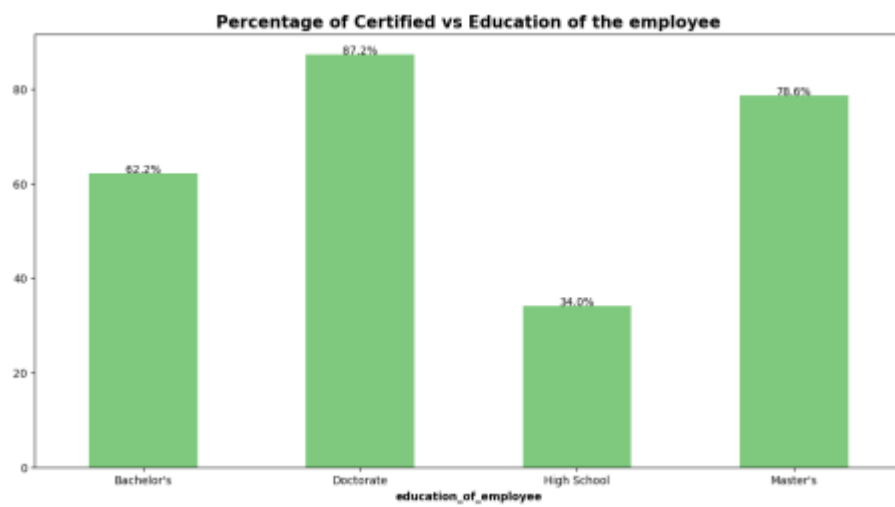
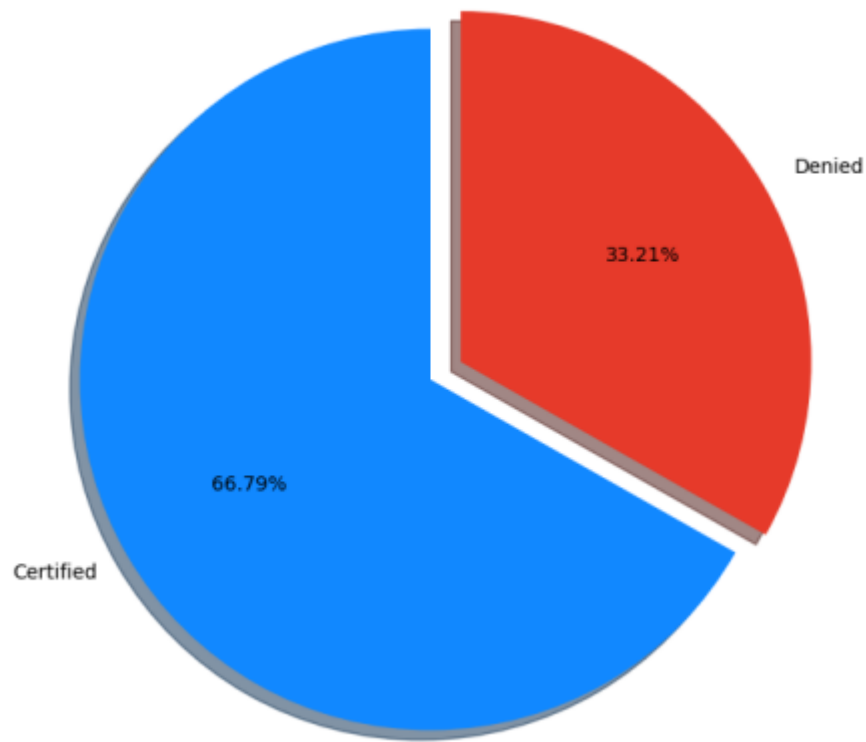


	Column	Hypothesis Result
0	continent	Reject Null Hypothesis
1	education_of_employee	Reject Null Hypothesis
2	has_job_experience	Reject Null Hypothesis
3	requires_job_training	Fail to Reject Null Hypothesis
4	region_of_employment	Reject Null Hypothesis
5	unit_of_wage	Reject Null Hypothesis
6	full_time_position	Reject Null Hypothesis
7	case_status	Reject Null Hypothesis

Report

Distribution of Numerical Features By Case Status

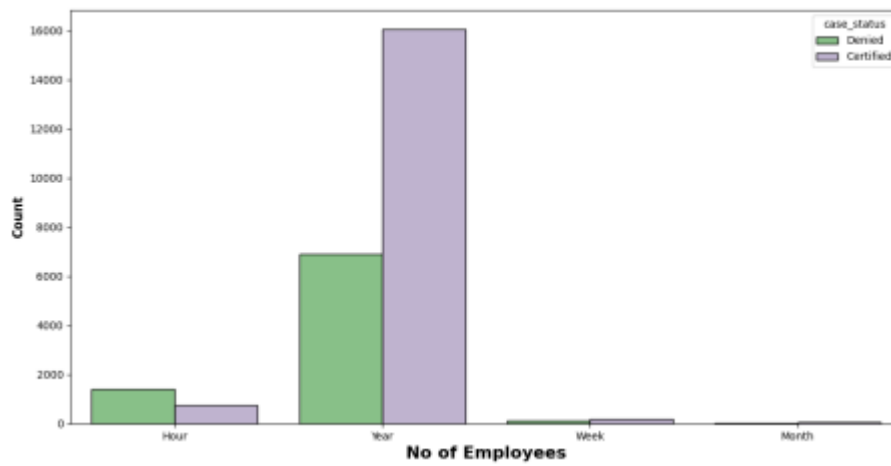




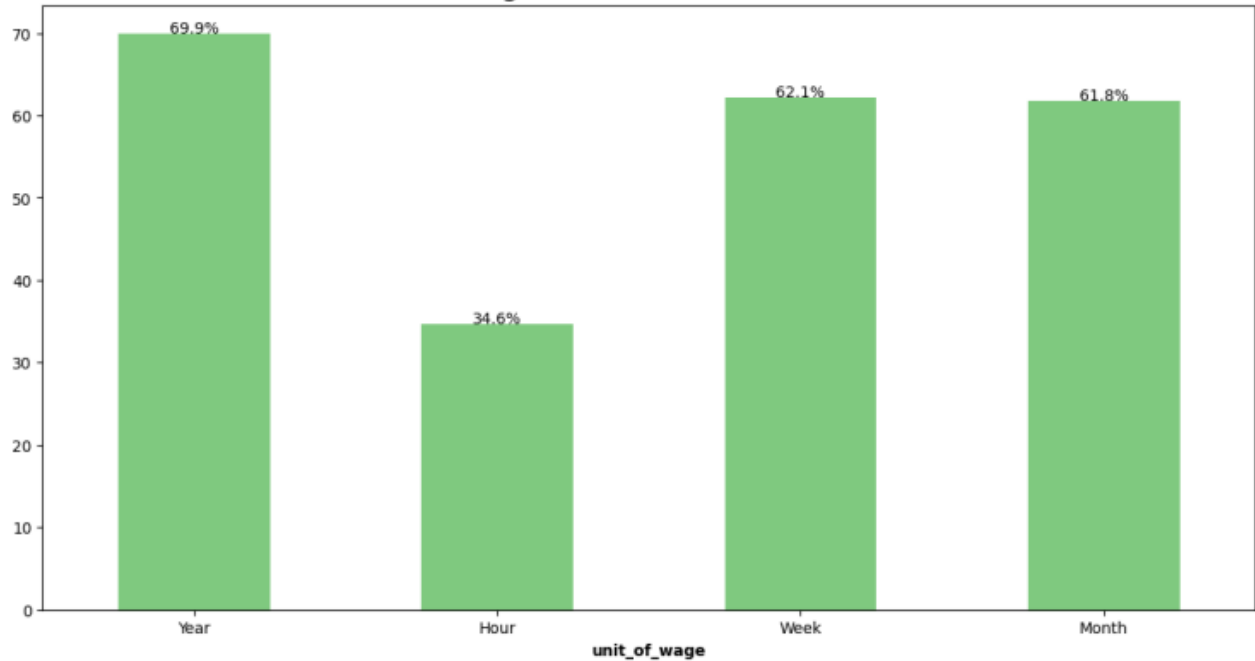
Requires job training for Employee vs Visa Status



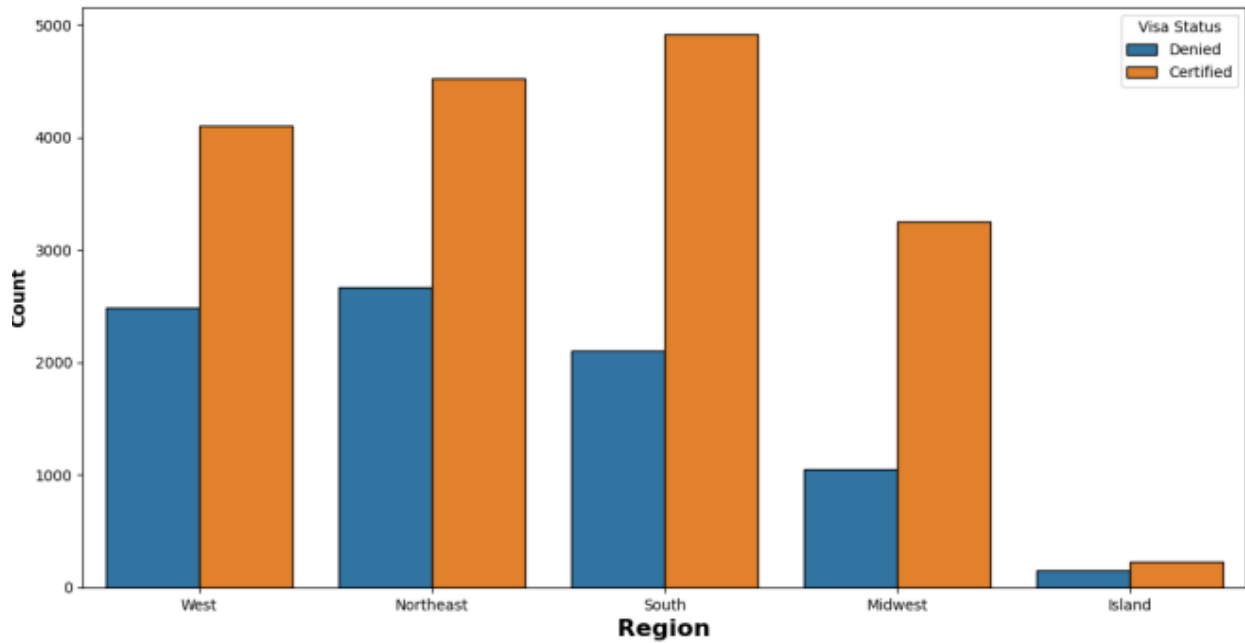
No of Employees in company vs Visa Status



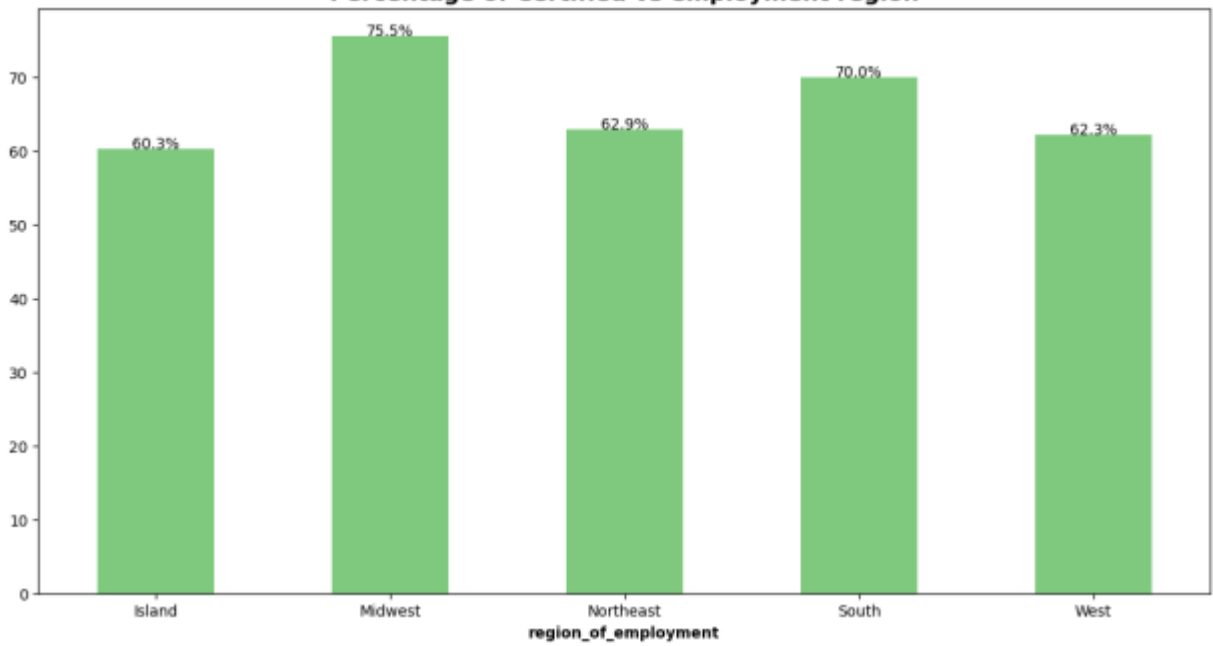
Percentage of Certified vs contract time



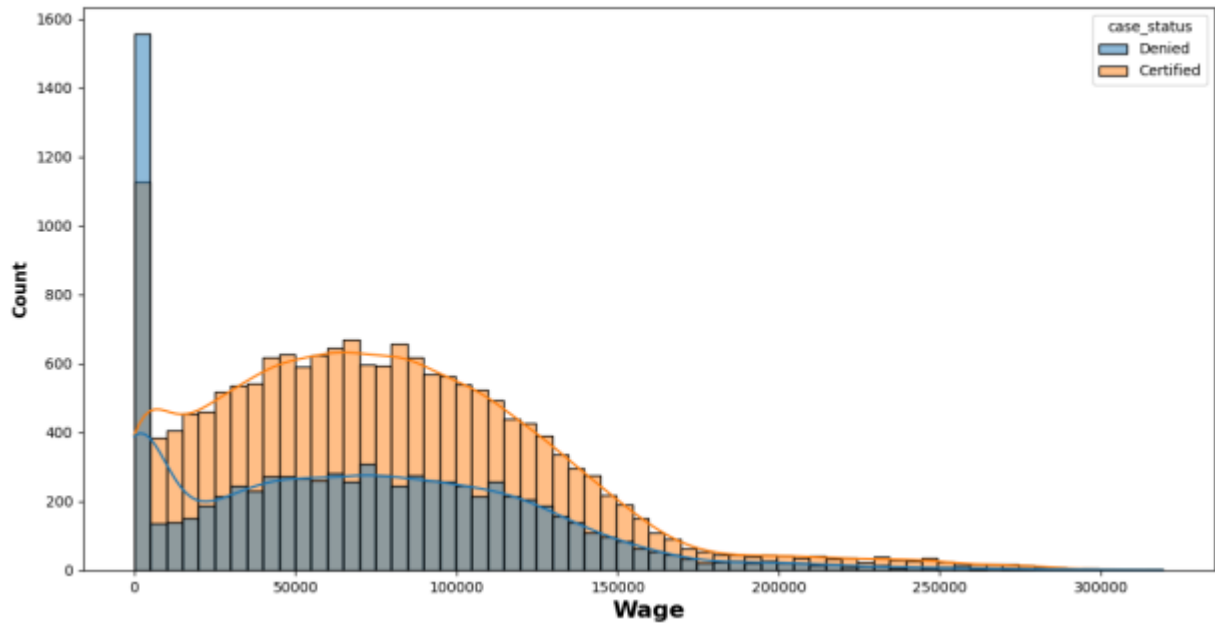
Region of Employment vs Visa Status



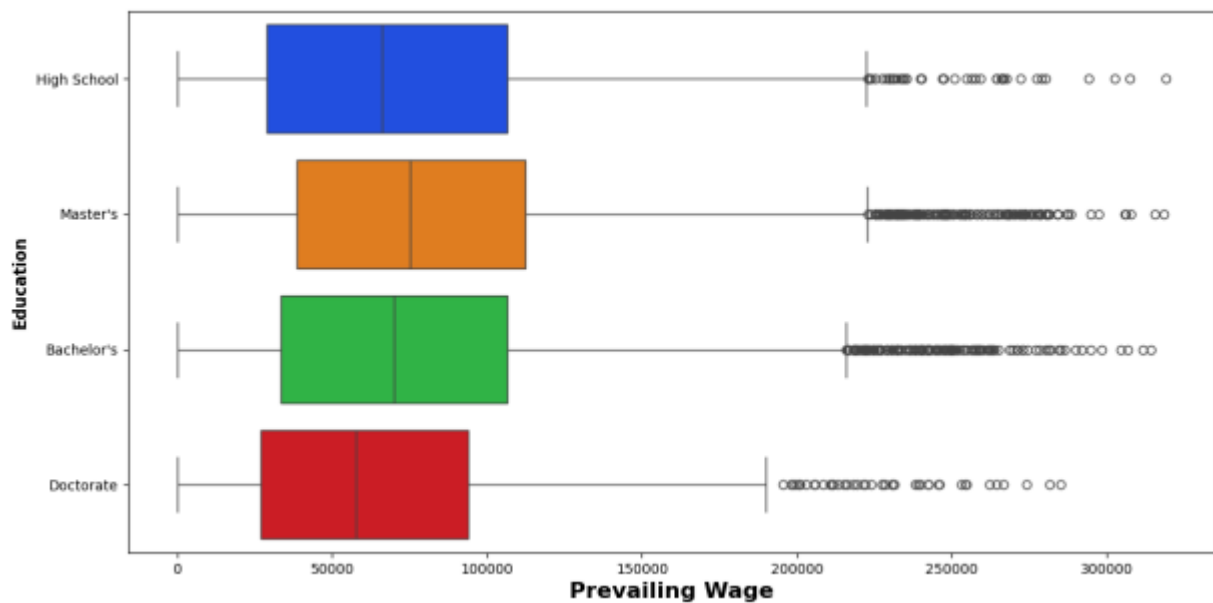
Percentage of Certified vs employment region



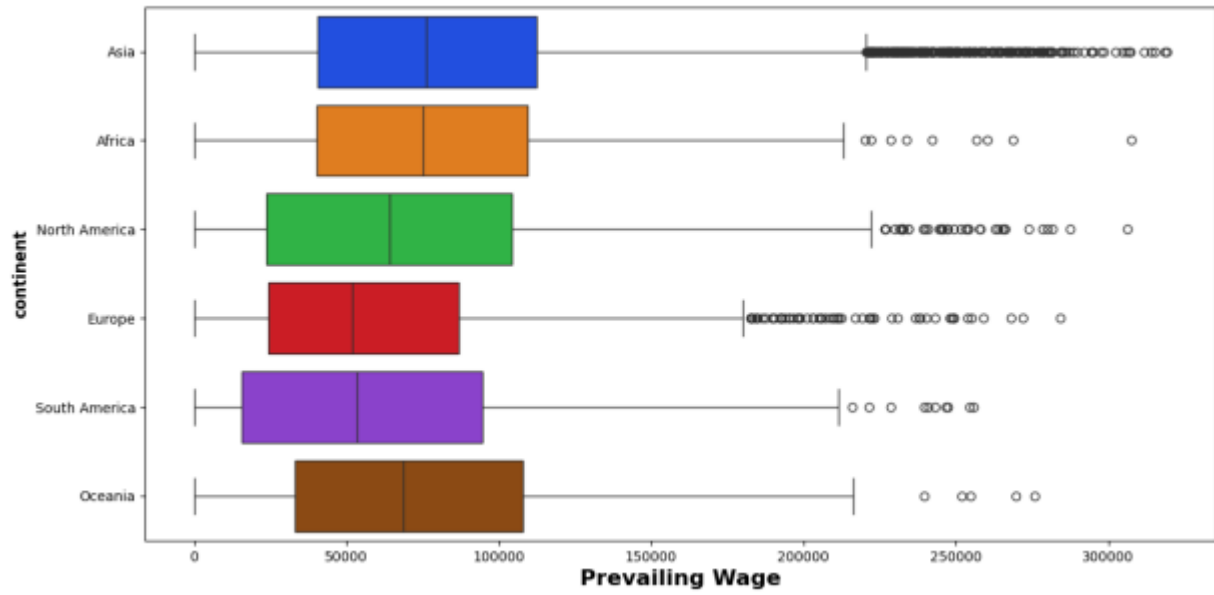
Prevailing wage vs Visa Status



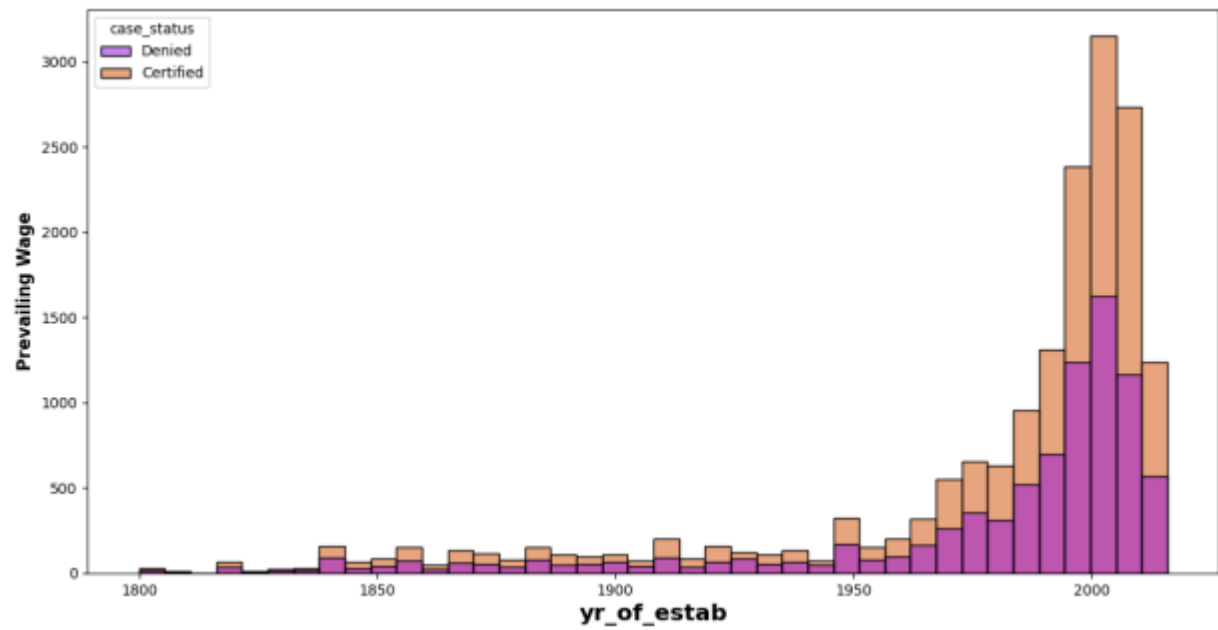
education_of_employee vs Prevailing Wage

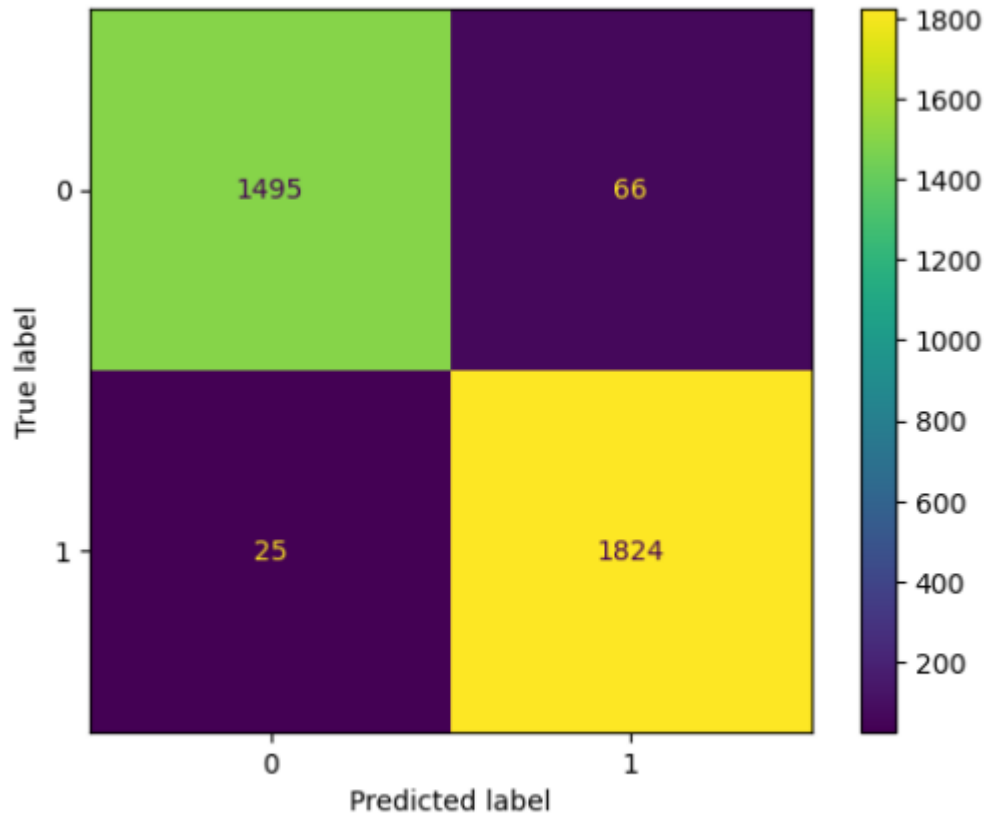


continent vs Visa Status



yr_of_estab vs Visa Status





Best Model is K-Nearest Neighbor(KNN) with Accuracy 97.33%

SCOPE

The scope of this report is to present the knowledge, methodologies, skills, and practical outcomes achieved through the completion of the **DevOps Fundamentals and Agile Methodologies** course by IBM and the **Machine Learning** course by SmartBridge in collaboration with Google for Developers. It focuses on understanding how modern software development practices and data-driven technologies can be applied to build efficient, scalable, and intelligent systems.

The scope includes the study of DevOps principles such as Continuous Integration and Continuous Delivery (CI/CD), automation of workflows, and the use of Agile methodologies like Scrum and Kanban for effective project management. It also covers the design and implementation of a CI/CD pipeline, demonstrating how development, testing, and deployment processes can be integrated and automated to improve productivity and software quality.

In addition, the scope extends to the field of Machine Learning, including data preprocessing, model development, evaluation, and optimization techniques. It highlights the development of a predictive model for H1-B visa approval, showcasing the application of machine learning algorithms to analyze real-world datasets and generate meaningful predictions. The report also considers the importance of data quality, feature selection, and performance evaluation in building reliable models.

Furthermore, the scope emphasizes the integration of both domains, illustrating how DevOps practices can support the deployment and maintenance of machine learning models in real-world environments. It provides an overview of how these combined technologies contribute to building robust, scalable, and data-driven applications aligned with current industry standards.

However, the scope is limited to the concepts, tools, and implementations covered within the duration of the courses. Advanced topics and large-scale industry implementations are beyond the scope of this report but provide opportunities for future exploration and learning.

CONCLUSION

In conclusion, the completion of the **DevOps Fundamentals and Agile Methodologies** course by IBM and the **Machine Learning** course by SmartBridge in collaboration with Google for Developers has provided a comprehensive understanding of both modern software development practices and data-driven technologies. These courses have not only strengthened theoretical knowledge but also enhanced practical skills through hands-on implementation of real-world projects.

Through the DevOps course, a clear understanding of automation, collaboration, and efficient software delivery was developed. The successful implementation of a CI/CD pipeline demonstrated the ability to integrate development, testing, and deployment processes into a streamlined workflow. This highlights the importance of speed, reliability, and continuous improvement in today's fast-paced software development environment.

Similarly, the Machine Learning course contributed to building strong analytical and problem-solving skills. The development of the H1-B visa prediction model showcased the practical application of machine learning concepts, including data preprocessing, model building, and performance evaluation. It reflected the ability to work with real datasets and derive meaningful insights to support decision-making.

Overall, the combination of these two domains has created a well-rounded skill set that bridges software engineering and intelligent systems. The knowledge gained from DevOps ensures efficient development and deployment, while machine learning adds the capability to build predictive and data-driven applications. This integrated learning experience prepares for future challenges in the field of information technology and provides a strong foundation for further growth, innovation, and professional development.

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